DC-DC CONVERTERS – INDUCTOR DESIGN (Magnetics)

MAGNETICS-1 Ferromagnetic Materials

To have a better understanding of the characteristics of magnetic-based devices (like inductors and transformers), it is important to have an understanding of the magnetic materials used in those devices.

At the most fundamental physics basis (classical), magnetic fields are created from moving electric charge. Often, we consider moving charge in the form of electric current.

However, we can also see that individual atoms have moving charge in the form of spinning electrons. Electron spin of unpaired electrons in an unfilled shell can result in a net magnetic field for the atom.

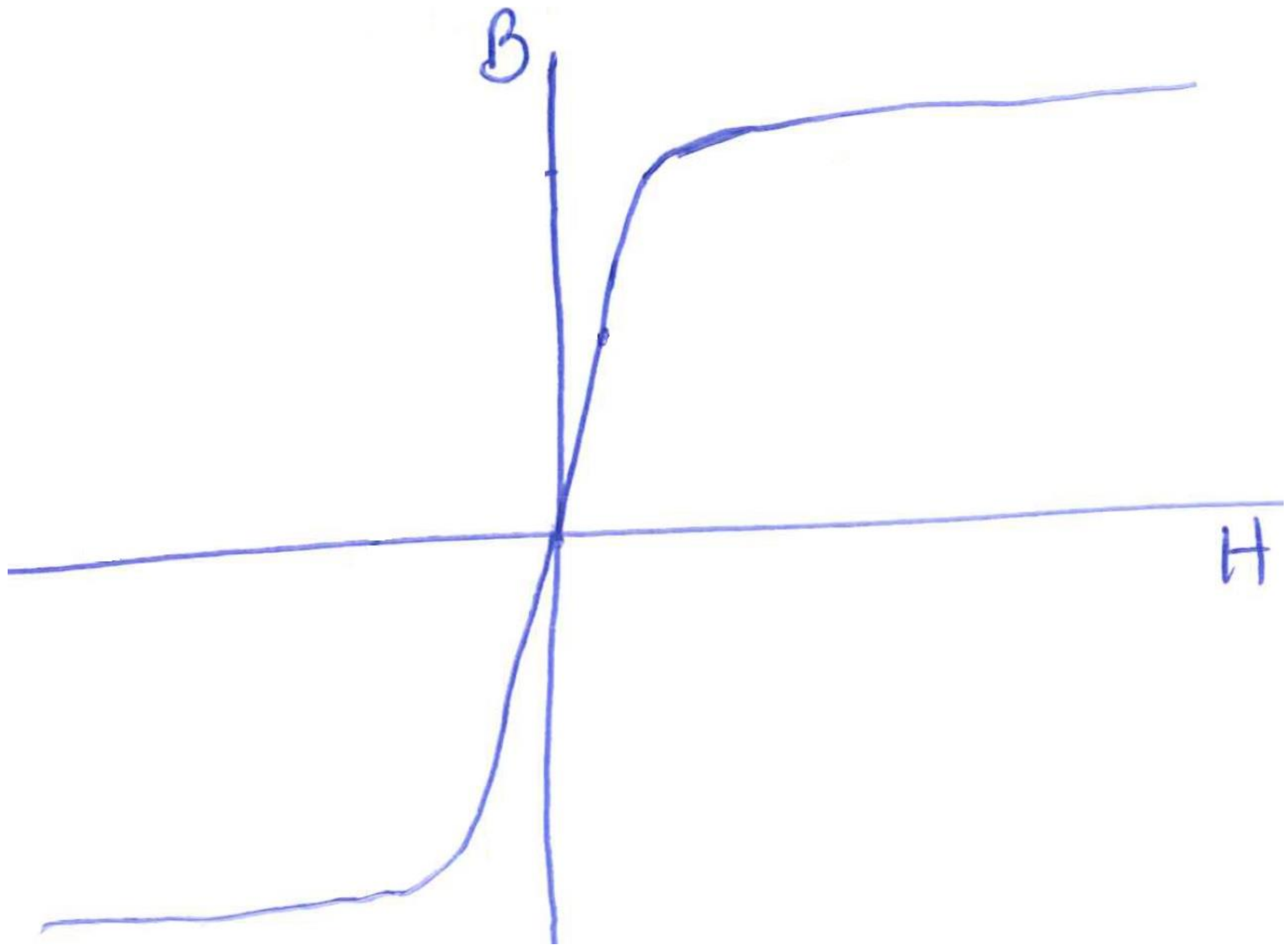
Iron has 4 unpaired electrons in its outer shell that all spin in the same direction.

So individual Iron atoms can be thought of as incredibly tiny permanent magnets.
What do permanent magnets do when you bring them into close proximity to each other?
The net magnetic flux from an individual Iron atom is strong enough to interact with other close Iron atom neighbours which results in a local magnetic force that causes alignment of the atoms and reinforcement of the overall magnetic flux.

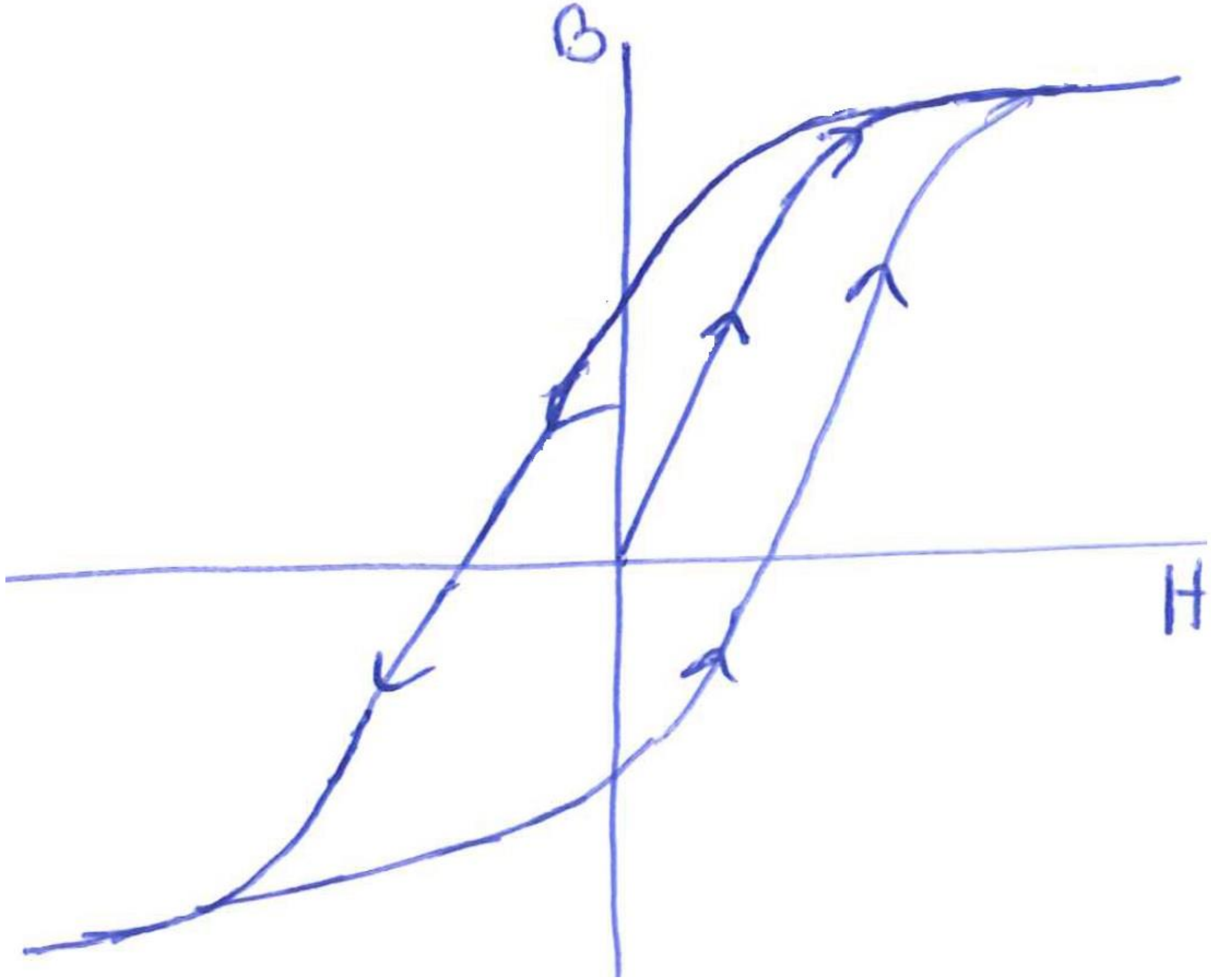
Physical production of bulk Iron-based molecule solid materials results in discontinuities such that the local magnetic flux reinforcement is bounded by small volume regions called "**domains**", and each domain is effectively a very small permanent magnet, normally randomly oriented. Domain magnetic flux direction can be manipulated though with an external magnetic force (field).

Alignment of magnetic domains increases the magnetising force by a property known as relative permeability (μ_r),

This effect is nicely illustrated via a B-H curve.

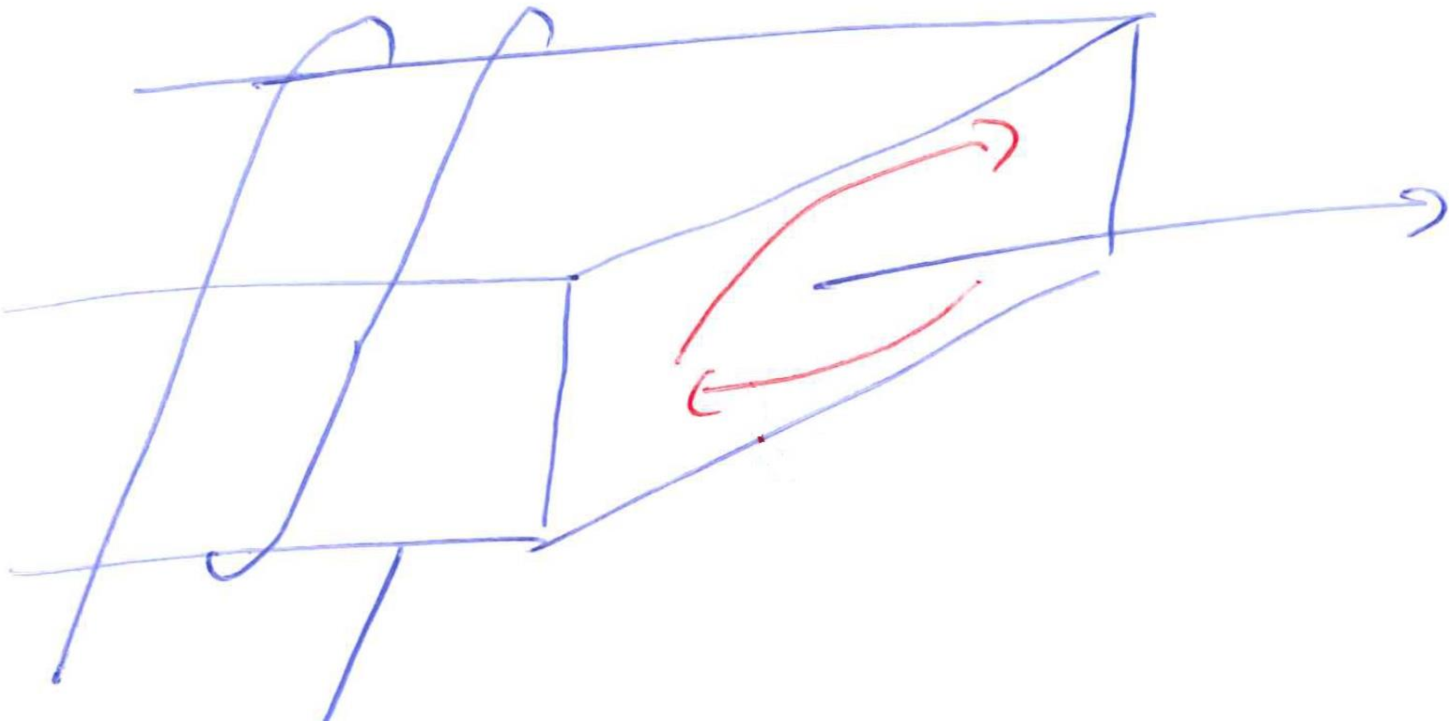


There is no power loss associated with a steady magnetic flux, but there is if it is changing.



Power loss scales with increasing “Hysteresis”.

We encounter another loss when the magnetic material also has finite conductivity. Consider;

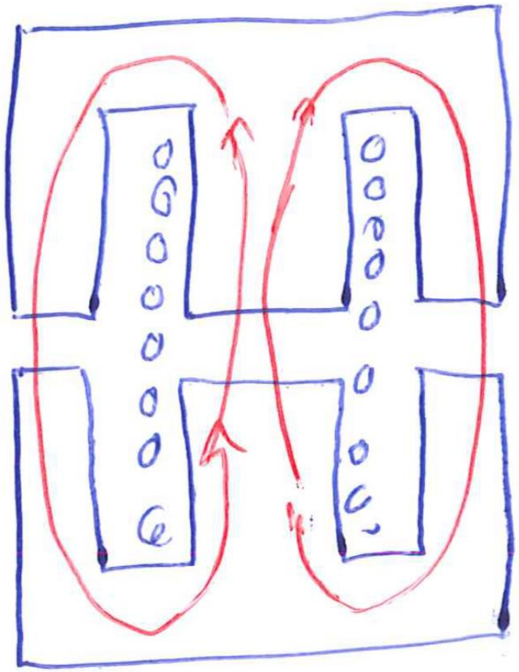


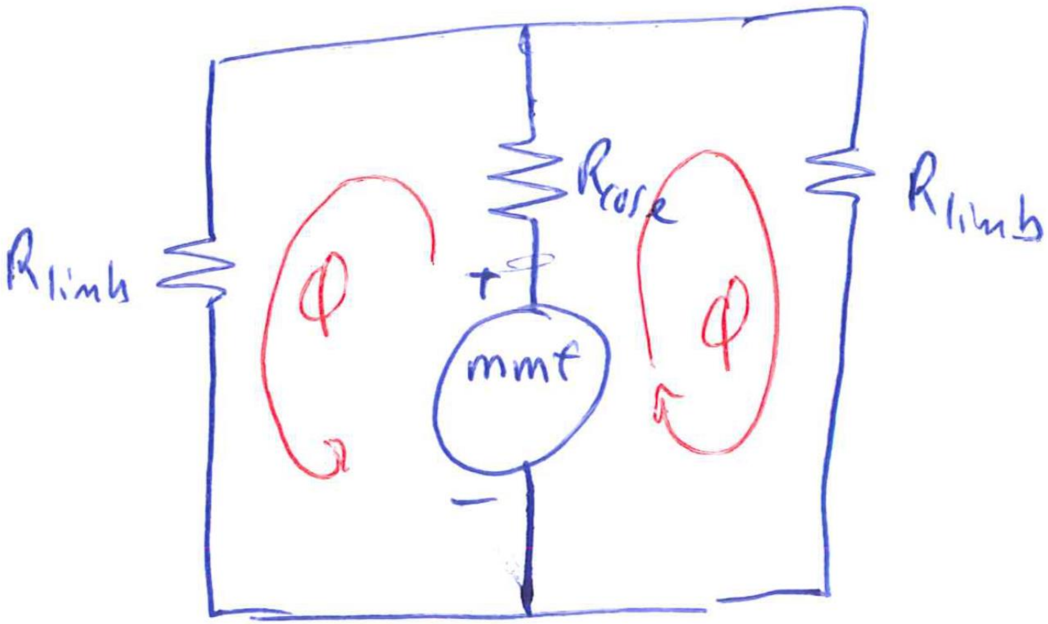
- Induced eddy current introduces $i_e^2 R$ power losses in the core.
- Opposes the change in magnetic flux.

MAGNETICS-2 Magnetic Circuits

Magnetic flux in a magnetic material core can be thought of very much like current flowing in a circuit.

Eg.



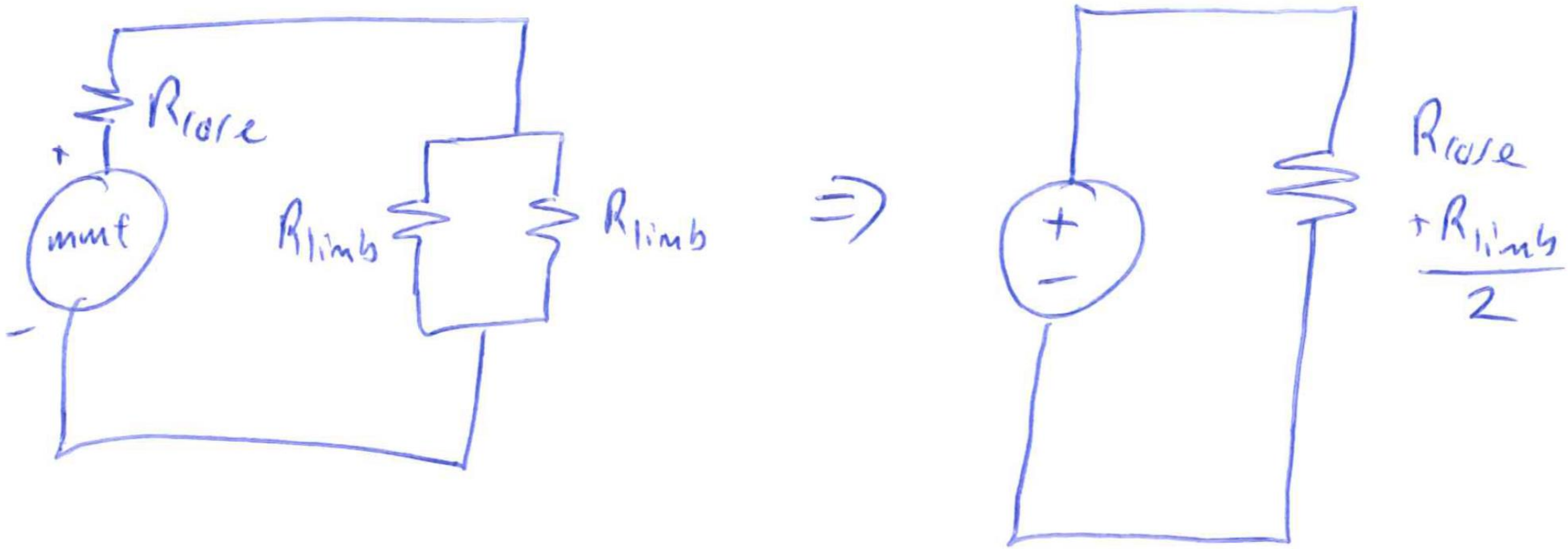


- Magnetising force: $mmf = Ni$ (analogous to voltage source in an electrical circuit)
- Reluctance: (analogous to resistance)

$$R = \frac{l}{A\mu_r\mu_0}$$

- Magnetic flux: $\phi = mmf/R$ (analogous to current)

Just like for electrical circuits with series and parallel-connected resistances, you can simplify magnetic circuits in exactly the same manner.



Need to consider the important effect of the small air gap in the core.

- Assume $\mu_r \approx 1600$ for ferrite and $\mu_r = 1$ for air. Given $R = \frac{l}{A\mu_r\mu_0}$, then the air gap reluctance dominates the total core reluctance.